MLP -

Forward Pass

In the hidden layer [i.e., j layer]

$$\begin{bmatrix} net_1 \\ \vdots \\ net_j \\ \vdots \\ net_m \end{bmatrix}_{m\times 1} = \begin{bmatrix} w_{10} & w_{11} & \cdots & w_{1i} & \cdots & w_{1d} \\ \vdots & \vdots & & \vdots & & \vdots \\ w_{j0} & w_{j1} & \cdots & w_{ji} & \cdots & w_{jd} \\ \vdots & & & & & \\ w_{m0} & w_{m1} & \cdots & w_{mi} & \cdots & w_{md} \end{bmatrix}_{m\times(d+1)} \begin{bmatrix} 1 \\ x_1 \\ \vdots \\ x_i \\ \vdots \\ x_d \end{bmatrix}_{(d+1)\times 1}$$

note, $x_0 = 1$

$$y_j = f(net_j), \quad 1 \le j \le m. \quad \boxed{y_0 = 1}$$

In the output layer [i.e., K layer]

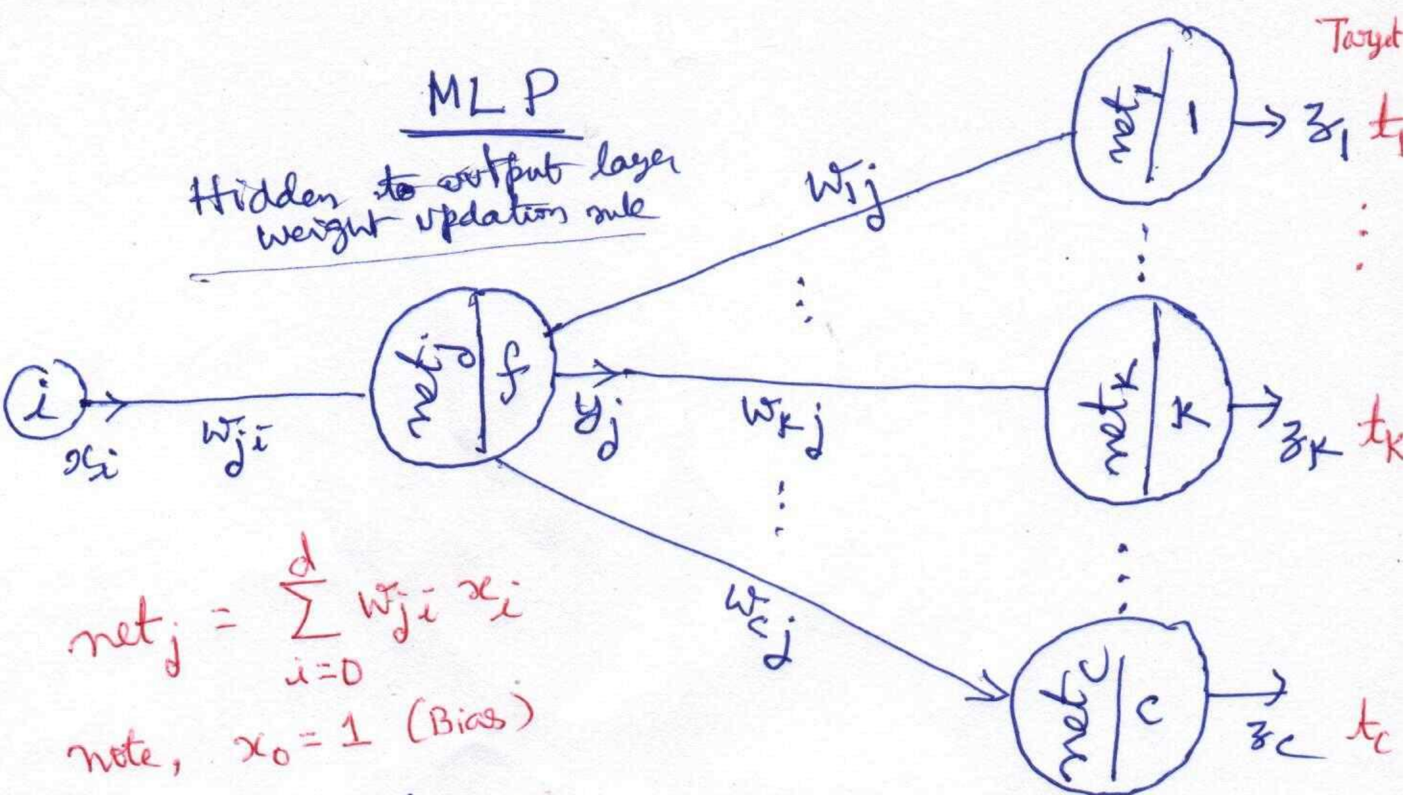
$$\begin{bmatrix} net_1 \\ \vdots \\ net_K \\ \vdots \\ net_c \end{bmatrix}_{c\times 1} = \begin{bmatrix} w_{10} & w_{11} & \cdots & w_{1j} & \cdots & w_{1m} \\ \vdots & \vdots & & & & \\ w_{K0} & w_{K1} & \cdots & w_{Kj} & \cdots & w_{Km} \\ \vdots & & & & & \\ w_{c0} & w_{c1} & \cdots & w_{cj} & \cdots & w_{cm} \end{bmatrix}_{c\times(m+1)} \begin{bmatrix} 1 \\ y_1 \\ \vdots \\ y_j \\ \vdots \\ y_m \end{bmatrix}_{(m+1)\times 1}$$

$$z_K = f(net_K), \quad 1 \le K \le c, \quad 0 \le j \le m,$$

$\boxed{\text{note, } y_0 = 1}$

MLP

Hidden to output layer weight updation rule



$$net_j = \sum_{i=0}^{d} w_{ji} x_i$$

note, $x_0 = 1$ (Bias)

$$J = \frac{1}{2} \sum_{r=1}^{c} (t_r - z_r)^2$$

$$net_k = \sum_{j=0}^{m} w_{kj} y_j$$

note, $y_0 = 1$ Bias

$$\frac{\partial J}{\partial w_{kj}} = \underbrace{\frac{\partial J}{\partial z_k} \cdot \frac{\partial z_k}{\partial net_k}}_{-\delta_k} \cdot \underbrace{\frac{\partial net_k}{\partial w_{kj}}}_{y_j}$$

$$= -(t_k - z_k) f'(net_k)\, y_j, \quad 1 \le k \le c, \; 0 \le j \le m$$

note, $y_0 = 1$

$$\Delta w_{kj} = \eta \, \delta_k \, y_j$$

input to hidden layer weight updation rule

$$\frac{\partial J}{\partial w_{ji}} = \underbrace{\frac{\partial J}{\partial net_j}}_{-\delta_j} \cdot \underbrace{\frac{\partial net_j}{\partial w_{ji}}}_{x_i}, \quad 0 \le i \le d, \quad 1 \le j \le m$$

net_0 is not defined

$y_0 = 1$

$$= \underbrace{\left(\sum_{r=1}^{c} \underbrace{\frac{\partial J}{\partial z_r} \cdot \frac{\partial z_r}{\partial net_r}}_{-\delta_r} \frac{\partial net_r}{\partial y_j}\right) \frac{\partial y_j}{\partial net_j}}_{-\delta_j} x_i$$

$$= \underbrace{\left(\sum_{r=1}^{c} \underbrace{-(t_r - z_r) f'(net_r)}_{-\delta_r} w_{rj}\right) f'(net_j)}_{-\delta_j} x_i$$

$0 \le i \le d, \quad 1 \le j \le m$

$x_0 = 1$

net_0 is undefined

$y_0 = 1$

$$\Delta w_{ji} = \eta \delta_j x_i$$

K - layer updation

$$\begin{bmatrix} \Delta w_{10} & \Delta w_{11} & \cdots & \Delta w_{1j} & \cdots & \Delta w_{1m} \\ \vdots & \vdots & & \vdots & & \vdots \\ \Delta w_{K0} & \Delta w_{K1} & \cdots & \Delta w_{Kj} & \cdots & \Delta w_{Km} \\ \vdots & & & & & \vdots \\ \Delta w_{c0} & \Delta w_{c1} & \cdots & \Delta w_{cj} & \cdots & \Delta w_{cm} \end{bmatrix}_{c \times (m+1)}$$

$$\cancel{= \eta \cdot \begin{matrix} y_0\delta_1 & y_1\delta_1 & \cdots & y_j\delta_1 & \cdots & y_m\delta_1 \\ \vdots & \vdots & & & & \\ y_0\delta_K & y_1\delta & & & & \end{matrix}}$$

$$= \eta \cdot \begin{bmatrix} \delta_1 y_0 & \delta_1 y_1 & \cdots & \delta_1 y_j & \cdots & \delta_1 y_m \\ \vdots & \vdots & & \vdots & & \vdots \\ \delta_K y_0 & \delta_K y_1 & \cdots & \delta_K y_j & \cdots & \delta_K y_m \\ \vdots & & & & & \\ \delta_c y_0 & \delta_c y_1 & \cdots & \delta_c y_j & \cdots & \delta_c y_m \end{bmatrix}_{c \times (m+1)}$$

$$= \eta \, \vec{\delta} \, \vec{Y}^t$$

where $\vec{\delta} = \begin{bmatrix} \delta_1 \\ \vdots \\ \delta_K \\ \vdots \\ \delta_c \end{bmatrix}_{c \times 1}$ $\quad \vec{Y} = \begin{bmatrix} y_0 \\ y_1 \\ \vdots \\ y_j \\ \vdots \\ y_m \end{bmatrix}_{(m+1) \times 1}$

we have, $\delta_K = (t_K - z_K) f'(net_K)$, $1 \le K \le c$.

j- layer updation

input to hidden layer updation rule

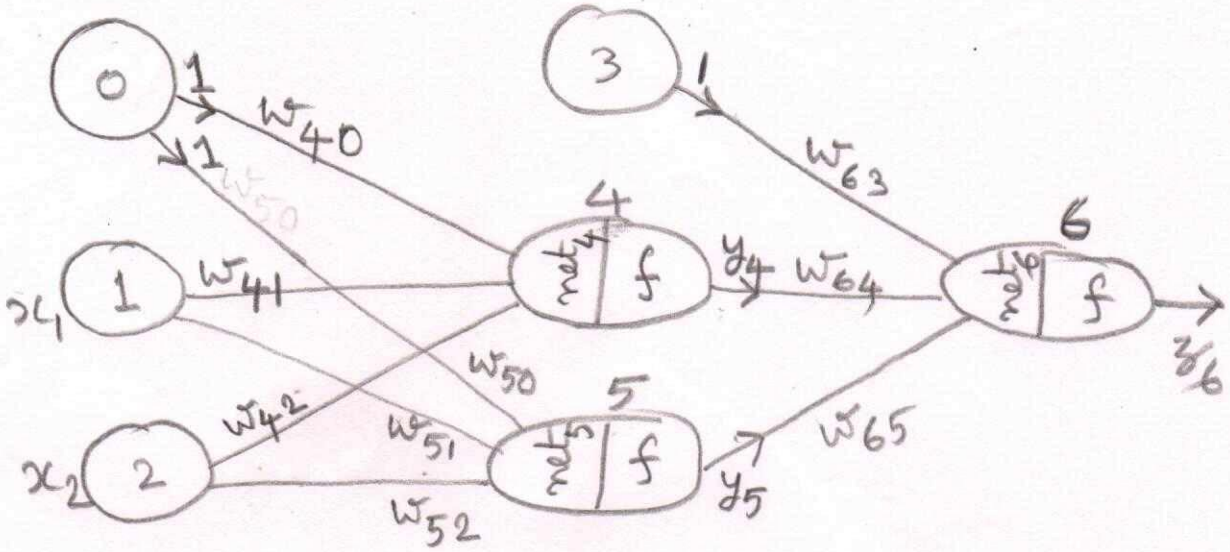
$$\begin{array}{cc} & i \\ j & \begin{bmatrix} \Delta w_{ji} \end{bmatrix} \end{array} = \eta \begin{bmatrix} \delta_1 \\ \delta_2 \\ \vdots \\ \delta_j \\ \vdots \\ \delta_m \end{bmatrix}_{m \times 1} \begin{bmatrix} x_0 & x_1 & \cdots & x_i & \cdots & x_d \end{bmatrix}_{1 \times (d+1)}$$

where

$$\delta_j = \left(\sum_{\gamma=1}^{c} \delta_\gamma w_{\gamma j} \right) f'(net_j)$$

$$1 \leq j \leq m, \quad 1 \leq \gamma \leq c$$

XOR



$$input = \begin{bmatrix} x_0 \\ x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \qquad target = t_6 = 1$$

Random initial Weights

$$\begin{bmatrix} w_{40} & w_{41} & w_{42} \\ w_{50} & w_{51} & w_{52} \end{bmatrix} = \begin{bmatrix} 0 & 0.1 & 0.1 \\ 0 & 0.2 & -0.1 \end{bmatrix}$$

$$\begin{bmatrix} w_{63} & w_{64} & w_{65} \end{bmatrix} = \begin{bmatrix} 0 & 0.1 & -0.1 \end{bmatrix}$$

Forward Pass

$$\begin{bmatrix} net_4 \\ net_5 \end{bmatrix} = \begin{bmatrix} 0 & 0.1 & 0.1 \\ 0 & 0.2 & -0.1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.1 \\ -0.1 \end{bmatrix}, \begin{bmatrix} y_4 \\ y_5 \end{bmatrix} = \begin{bmatrix} 0.525 \\ 0.475 \end{bmatrix}$$

$$[net_6] = \begin{bmatrix} 0 & 0.1 & -0.1 \end{bmatrix} \begin{bmatrix} 1 \\ 0.525 \\ 0.475 \end{bmatrix} = 0.005, \; z_6 = 0.501$$

$$\delta_6 = (t_6 - z_6)\, f'(net_6) = (1-0.501)(0.245)$$
$$= 0.122$$

$$[\Delta w_{63}\ \Delta w_{64}\ \Delta w_{65}] = \eta\, \delta_6\, [1\ y_4\ y_5]$$
$$= 1(0.122)[1\ 0.525\ 0.475]$$
$$= [0.122\ 0.064\ 0.058]$$

For simplicity, let $\eta = 1$

So, $$[w_{63}\ w_{64}\ w_{65}] = [0\ 0.1\ -0.1] + [0.122\ 0.064\ 0.058]$$
$$= [0.122\ 0.164\ -0.042]$$

$$\begin{bmatrix}\delta_4 \\ \delta_5\end{bmatrix} = \begin{bmatrix}\delta_6 \cdot w_{64} \cdot f'(net_4) \\ \delta_6 \cdot w_{65} \cdot f'(net_5)\end{bmatrix} = \begin{bmatrix}(0.122)(0.164)(0.249) \\ (0.122)(-0.042)(0.249)\end{bmatrix}$$
$$= \begin{bmatrix}0.005 \\ -0.001\end{bmatrix}$$

$$\begin{bmatrix}\Delta w_{40} & \Delta w_{41} & \Delta w_{42} \\ \Delta w_{50} & \Delta w_{51} & \Delta w_{52}\end{bmatrix} = \eta \begin{bmatrix}\delta_4 \\ \delta_5\end{bmatrix}[1\ x_1\ x_2] = \begin{bmatrix}0.005 \\ -0.001\end{bmatrix}[1\ 0\ 1]$$

we took $\eta = 1$

$$= \begin{bmatrix}0.005 & 0 & 0.005 \\ -0.001 & 0 & -0.001\end{bmatrix}$$

$$\begin{bmatrix}w_{40} & w_{41} & w_{42} \\ w_{50} & w_{51} & w_{52}\end{bmatrix} = \begin{bmatrix}0 & 0.1 & 0.1 \\ 0 & 0.2 & -0.1\end{bmatrix} + \begin{bmatrix}0.005 & 0 & 0.005 \\ -0.001 & 0 & -0.001\end{bmatrix}$$
$$= \begin{bmatrix}0.005 & 0.1 & 0.105 \\ -0.001 & 0.2 & -0.101\end{bmatrix}$$

XOR using MLP (Final answer)